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Module - 2

Chunking, Embeddings, VectorDB & Tools

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Topics to be Covered

- Data Loaders
- Splitters
- Embeddings Model
- Vector Database
- Tools

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RAG on Custom Knowledge Base

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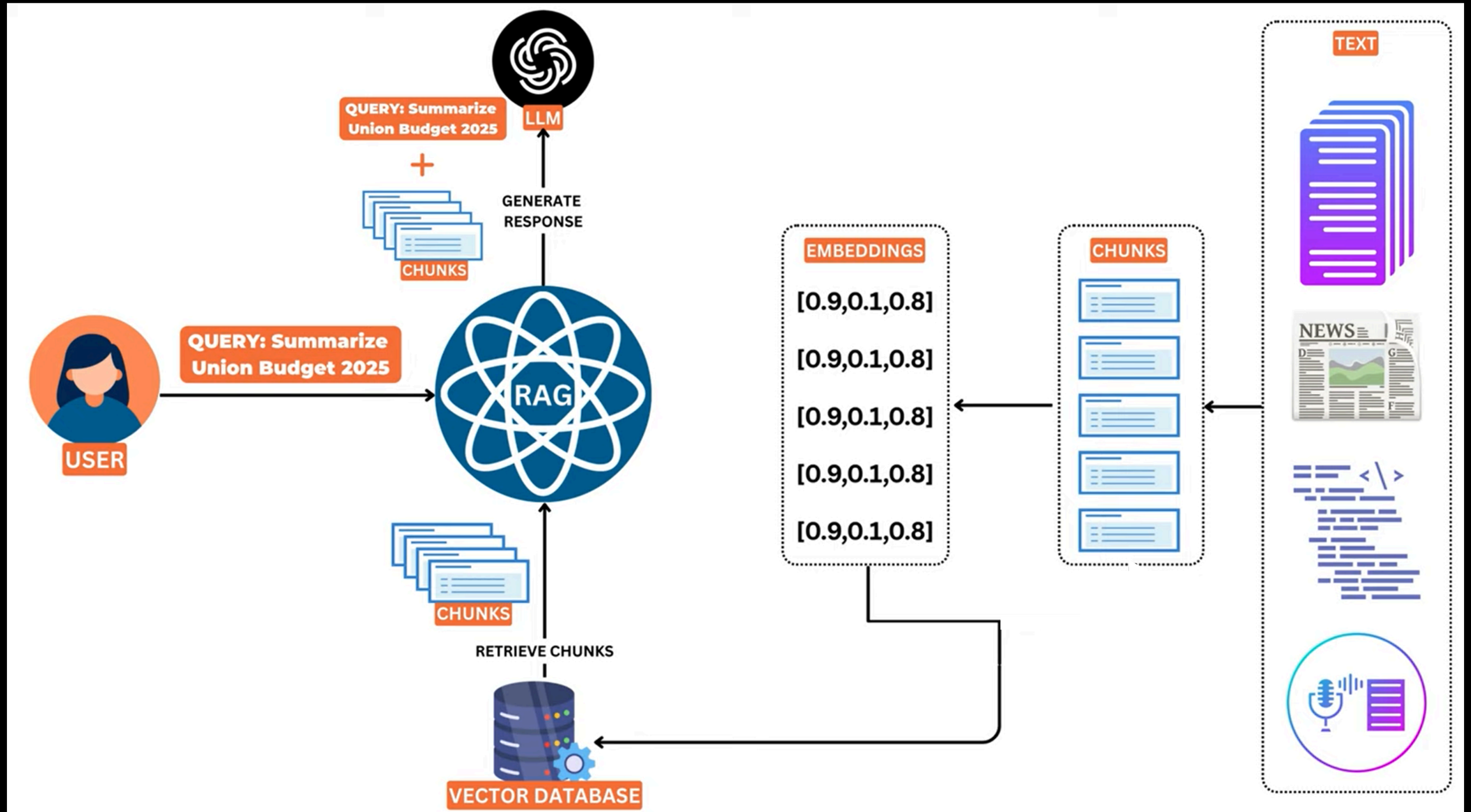
RAG Architecture



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Data Loaders

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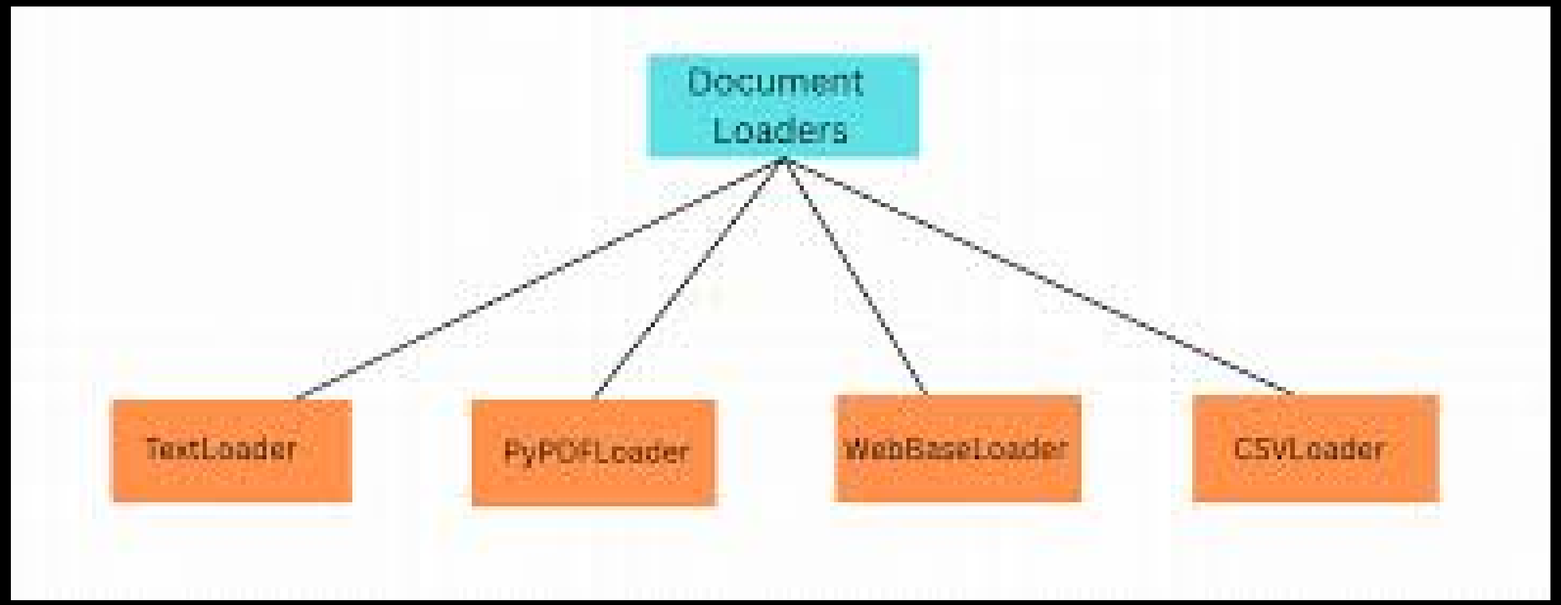
What is Data Loader?

In LangChain, It is a component to load the data.



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Text Loader

```
from langchain_community.document_loaders import TextLoader
loader = TextLoader("few_shot_prompt.txt")

documents = loader.load()
for doc in documents:
    print(f"Content: {doc.page_content}")
    print(f"Metadata: {doc.metadata}")
```

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PDF Loader

```
# PDF Loader
from langchain_community.document_loaders import PyPDFLoader
loader = PyPDFLoader("./docs/IndustrialAI_Project.pdf")

pages = loader.load()
for p in pages:
    print(f"Content: {p.page_content}")
    print(f"Metadata: {p.metadata}")
```

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Upload A Directory – PDF

```
# Directory Loader
from langchain_community.document_loaders import DirectoryLoader, PyPDFLoader

loader = DirectoryLoader("./docs", glob="*.pdf", loader_cls=PyPDFLoader)

docs = loader.load()
for p in pages:
    print(f"Content: {p.page_content}")
    print(f"Metadata: {p.metadata}")
```

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Upload A Directory – Markdown → 1

```
# Directory Loader
from langchain_community.document_loaders import (
    DirectoryLoader,
    UnstructuredMarkdownLoader
)

loader = DirectoryLoader(
    path="./markdown_docs",
    glob="**/*.md",
    loader_cls=UnstructuredMarkdownLoader,
    recursive=True # Set to True to load files from subdirectories as well
)
```

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Upload A Directory – Markdown → 2

```
docs = loader.load()
print(f"Loaded {len(docs)} documents.")
if docs:
    print("\nSnippet of the first document:")
    print(docs[0].page_content[:500])
    print("\nMetadata of the first document:")
    print(docs[0].metadata)
```

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Chunking

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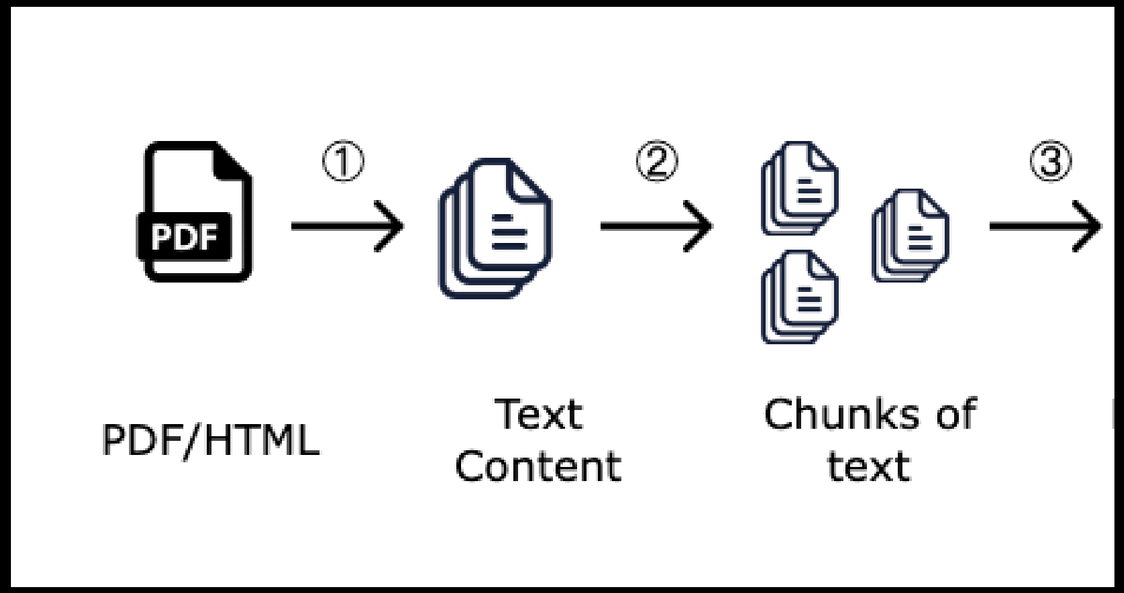
What is Chunking?

Chunking means breaking a large piece of text into smaller, manageable pieces called chunks.



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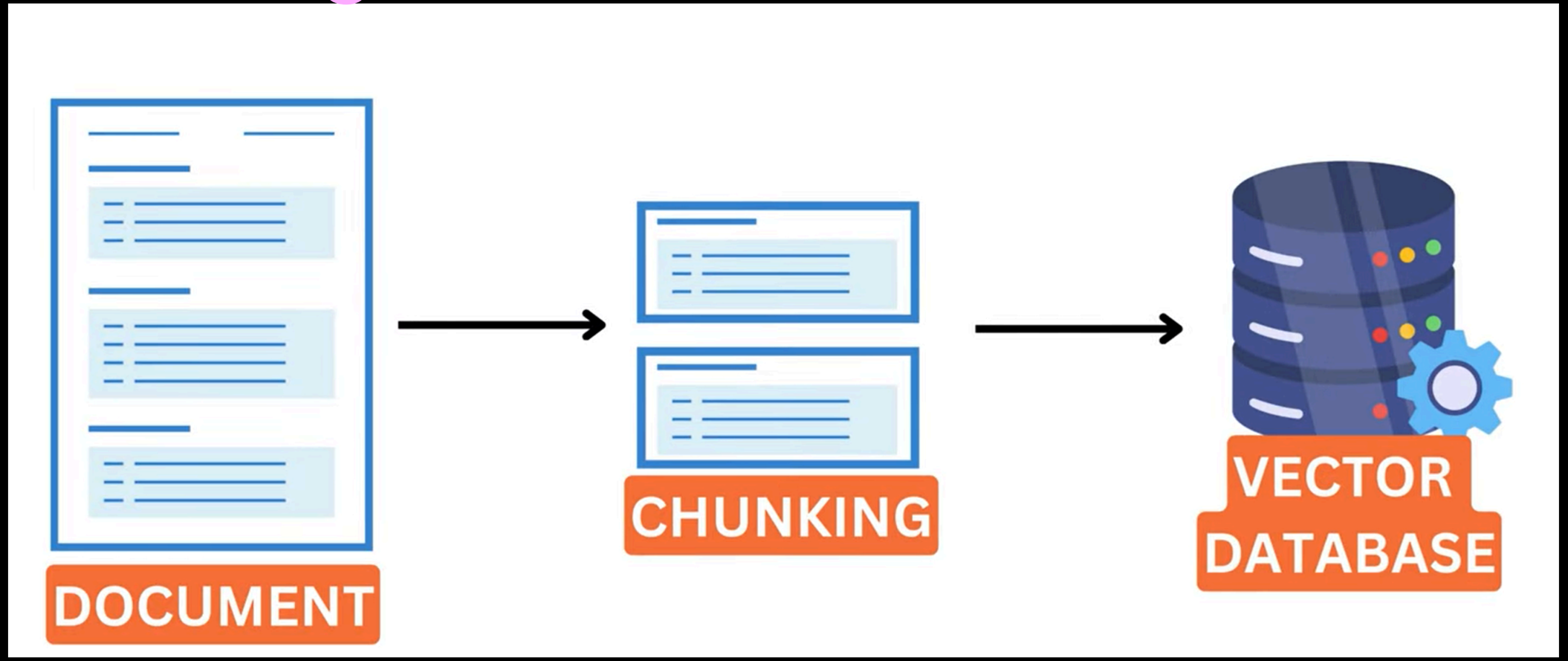
Chunking Visualization



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Why Chunking?

Following are the reasons due to we implement chunking:

- More relevant retrieval
- Better LLM context

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Case Study: Impact of Improper Chunking

A fintech company nearly lost a major deal because their AI chatbot provided incorrect legal advice. When asked about indemnification clauses in an NDA, their system confidently stated that “party A fully indemnifies party B” — omitting the critical exception clause that appeared in the next chunk. The contract actually read “party A indemnifies party B, except as provided in section 12.3,” but a poor chunking strategy had split this sentence across two separate pieces.

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Chunking Strategies

Following are the chunking strategies that we will explore:

- Text-based Chunking
- Document-based Chunking
- Contextual Chunking by Anthropic

To Learn More About Chunking Strategies (Splitters) in Langchain Visit:
https://docs.langchain.com/oss/python/integrations/splitters

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Text-based Chunking → RecursiveCharacterTextSplitter

```python
from langchain_text_splitters import RecursiveCharacterTextSplitter

# Create a recursive text splitter
recursive_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,                    # Target chunk size in characters
    chunk_overlap=100,                 # Overlap between chunks
    separators=[
        "\n\n",      # First, try to split on double newlines (paragraphs)
        "\n",        # Then single newlines
        ". ",        # Then sentences
        ", ",        # Then clauses
        " ",         # Then words
        ""           # Finally, characters (last resort)
    ]
)
```

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Text-based Chunking → RecursiveCharacterTextSplitter

```
# Sample structured document
structured_document = """
Place your text to chunk here
"""
# Split the document recursively
recursive_chunks = recursive_splitter.split_text(structured_document)

# Display the chunks with their characteristics
for i, chunk in enumerate(recursive_chunks):
    print(f"Chunk {i + 1} (length: {len(chunk)} chars):")
    print(chunk)
    print("-" * 60)
```

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Document-based Chunking → Language Splitter

```
from langchain_text_splitters import (
    Language,
    RecursiveCharacterTextSplitter,
)

PYTHON_CODE = """
def hello_world():
    print("Hello, World!")

# Call the function
hello_world()
"""
```

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Document-based Chunking → Language Splitter

```
# Default Seperators
# ['\nclass ', '\ndef ', '\n\tdef ', '\n\n', '\n', ' ', '']

python_splitter = RecursiveCharacterTextSplitter.from_language(
    language=Language.PYTHON, chunk_size=50, chunk_overlap=0
)
python_docs = python_splitter.create_documents([PYTHON_CODE])
print(python_docs)
```

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Embeddings

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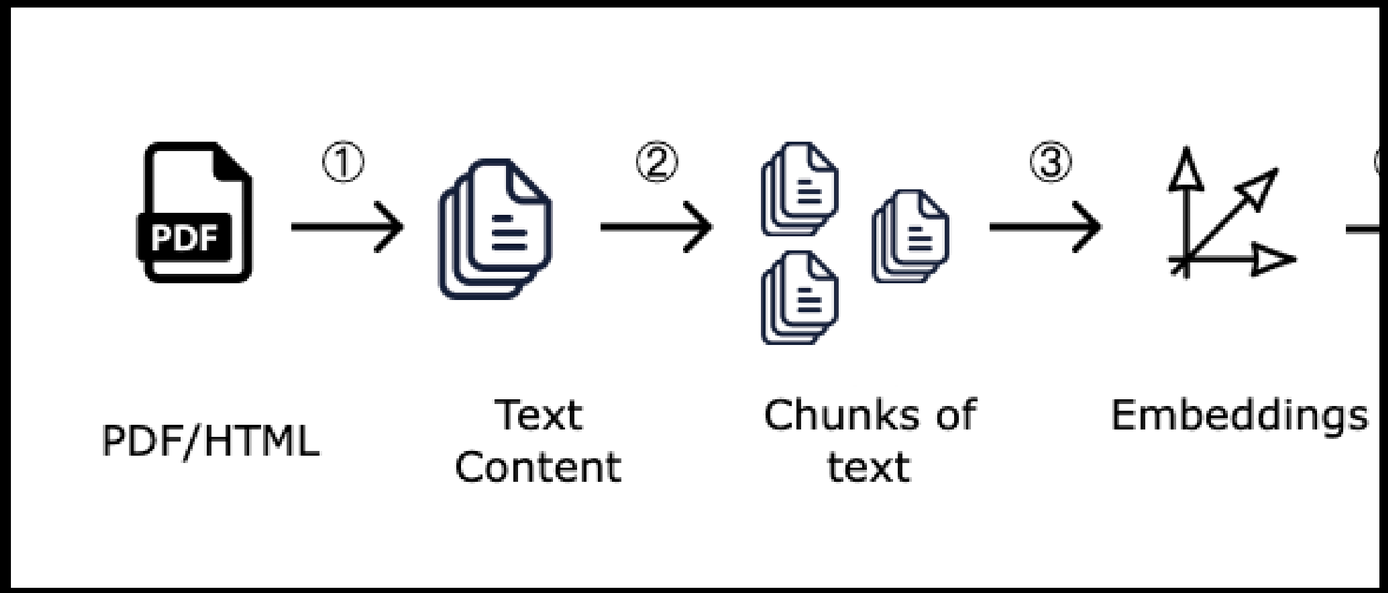
What are Embeddings?

Embeddings are numerical representations of text (or other data like images, audio, etc.) in a high-dimensional vector space.



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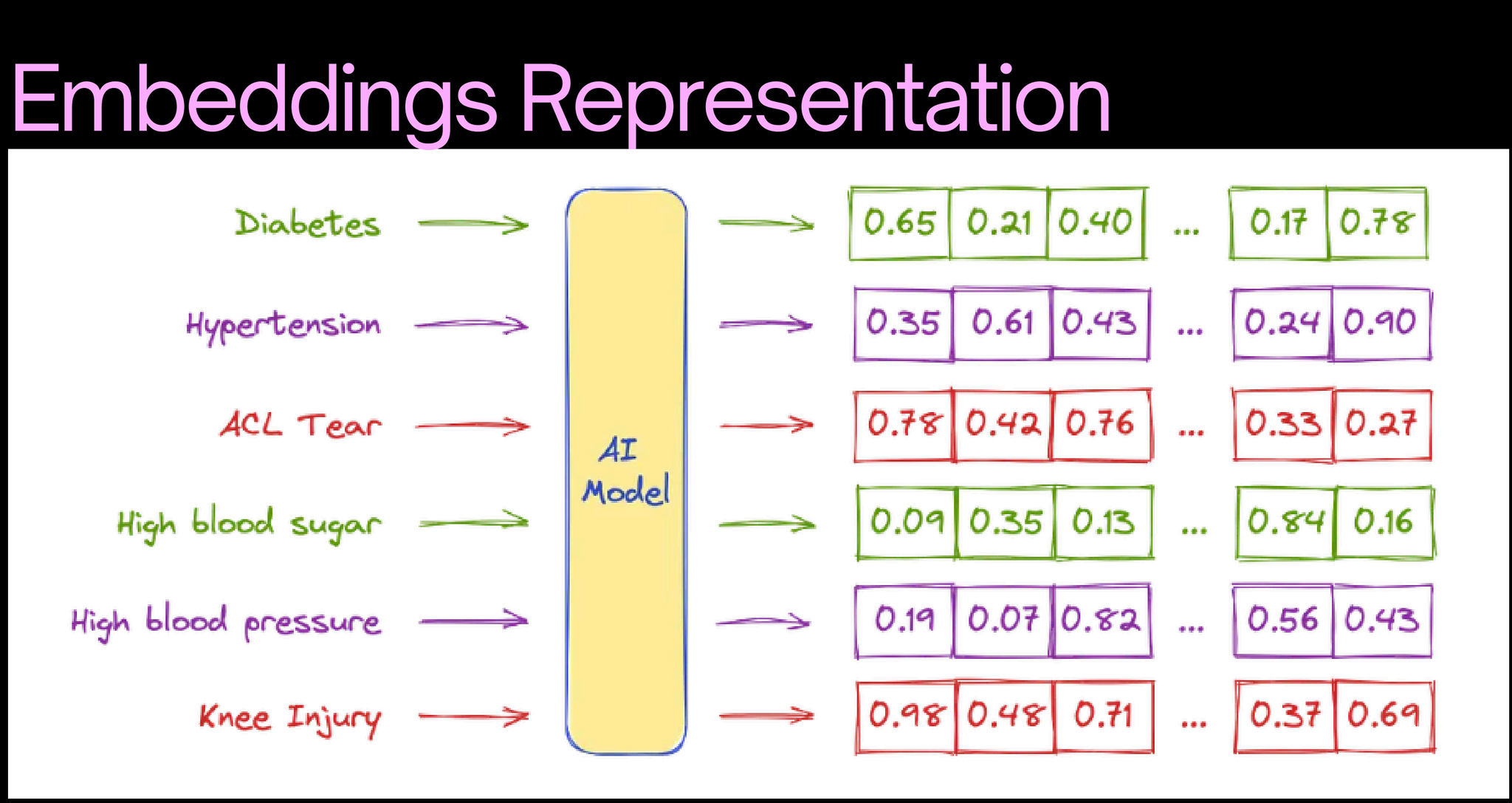
Embeddings Representation



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Embeddings Model

We will be using following embeddings model:

- text-embedding-3-large (Max Tokens: 8191)
- all-MiniLM-L6-v2 (Max Tokens: 256)

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For More Embeddings Model

Visit LeaderBoard: https://huggingface.co/spaces/mteb/leaderboard

Summary | Performance per Model Size | Performance over Time | Performance per Task Type | Performance per task | Performance per language | Task information

Filter...

Rank (Bor…	Model	Embedding D…	Max Tokens	Mean (T…	Mean (TaskT…	Bitext …	Classification	Clustering	Instruction R…	Multilabel Cl…
1	KaLM-Embedding-Gemma3-12B-2511	3840	32768	**72.32**	**62.51**	**83.76**	**77.88**	55.77	5.49	33.03
2	llama-embed-nemotron-8b	4096	32768	69.46	61.09	81.72	73.21	54.35	10.82	29.86
3	Qwen3-Embedding-8B	4096	32768	70.58	61.69	80.89	74.00	57.65	10.06	28.66
4	gemini-embedding-001	3072	2048	68.37	59.59	79.28	71.82	54.59	5.18	29.16
5	Qwen3-Embedding-4B	2560	32768	69.45	60.86	79.36	72.33	57.15	11.56	26.77
6	Octen-Embedding-8B	4096	32768	67.84	60.28	80.35	66.68	55.68	8.90	25.23
7	F2LLM-v2-14B	5120	40960	68.74	59.45	77.15	73.00	**60.91**	0.62	28.14
8	F2LLM-v2-8B	4096	40960	68.09	58.99	75.96	71.93	60.62	0.85	27.38
9	Seed1.6-embedding-1215	2048	32768	70.26	61.34	78.68	76.75	56.78	-0.02	**46.16**
10	F2LLM-v2-4B	2560	40960	67.06	58.25	74.49	70.73	59.53	2.25	26.58
11	jina-embeddings-v5-text-small	1024	32768	67.00	58.90	69.71	71.32	53.41	1.35	41.97

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OpenAI Embeddings Model: text-embedding-3-large

For Single Query Embeddings

```python
from langchain_openai import OpenAIEmbeddings
from dotenv import load_dotenv

load_dotenv()

embedding = OpenAIEmbeddings(model='text-embedding-3-large', dimensions=5)

result = embedding.embed_query("When Pakistan became nuclear power?")

print(result)
```

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OpenAI Embeddings Model: text-embedding-3-large

For Multiple Documents Embeddings

```python
from langchain_openai import OpenAIEmbeddings
from dotenv import load_dotenv
load_dotenv()

embedding = OpenAIEmbeddings(model='text-embedding-3-large', dimensions=5)
documents = [
    "Islamabad is the capital of Pakistan",
    "Karachi is the largest city of Pakistan",
    "Lahore is famous for its cultural heritage"
]

result = embedding.embed_documents(documents)
print(result)
```

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SentanceTransformers Embeddings Model: all-mpnet-base-v2

For Single Query Embeddings

```
from langchain_huggingface import HuggingFaceEmbeddings

embedding = HuggingFaceEmbeddings(model_name='sentence-transformers/all-MiniLM-L6-v2')

result = embedding.embed_query("When Pakistan became nuclear power?")

print(result)
```

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SentanceTransformers Embeddings Model: all-mpnet-base-v2

For Multiple Documents Embeddings

```python
from langchain_huggingface import HuggingFaceEmbeddings

embedding = HuggingFaceEmbeddings(model_name='sentence-transformers/all-MiniLM-L6-v2')

documents = [
    "Islamabad is the capital of Pakistan",
    "Karachi is the largest city of Pakistan",
    "Lahore is famous for its cultural heritage"
]

vector = embedding.embed_documents(documents)

print(vector)
```

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Vector Database

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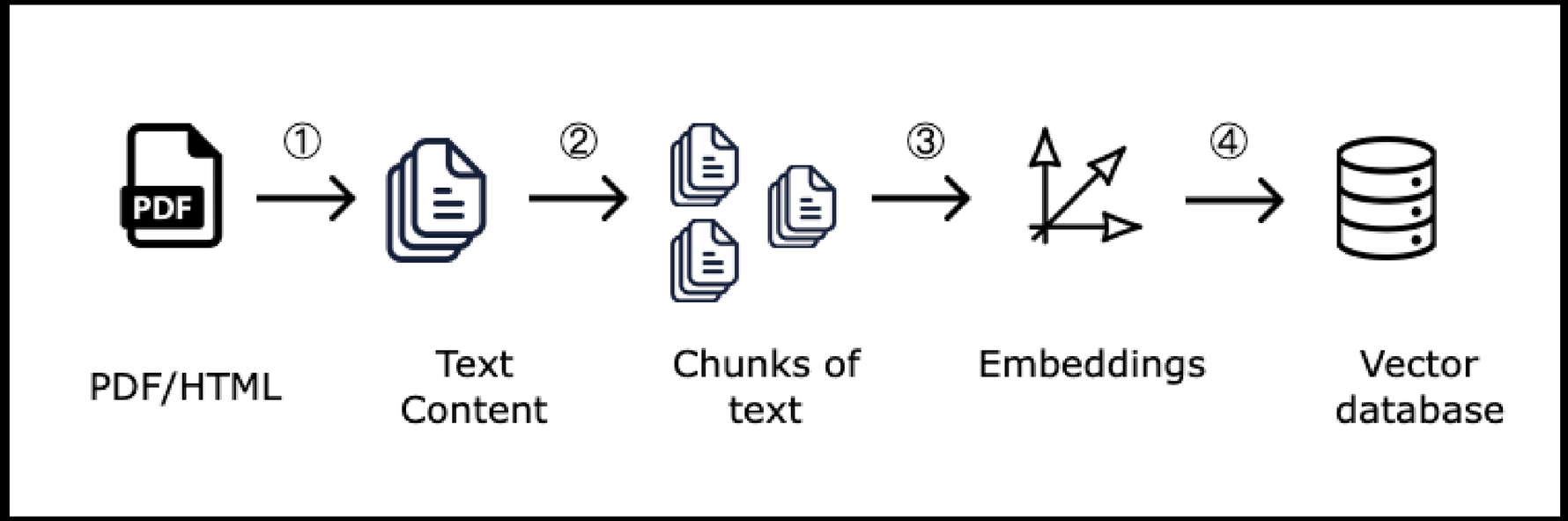
What is VectorDB?

A Vector Database (Vector DB) is a database optimized for storing and querying vector embeddings instead of traditional rows/columns.



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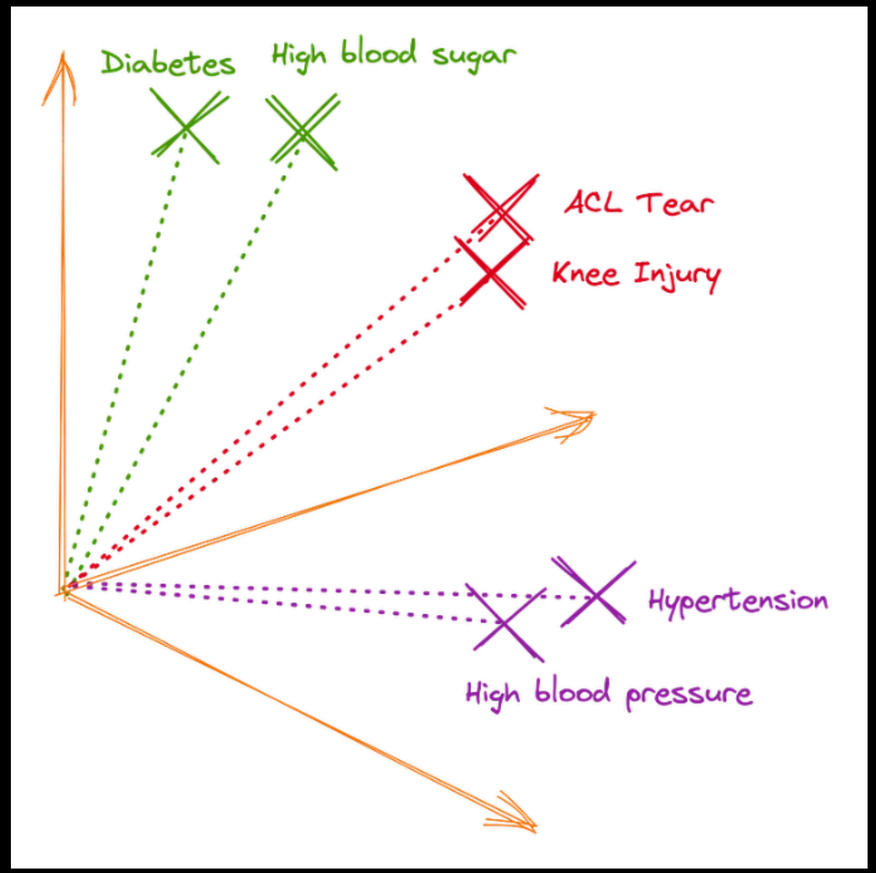
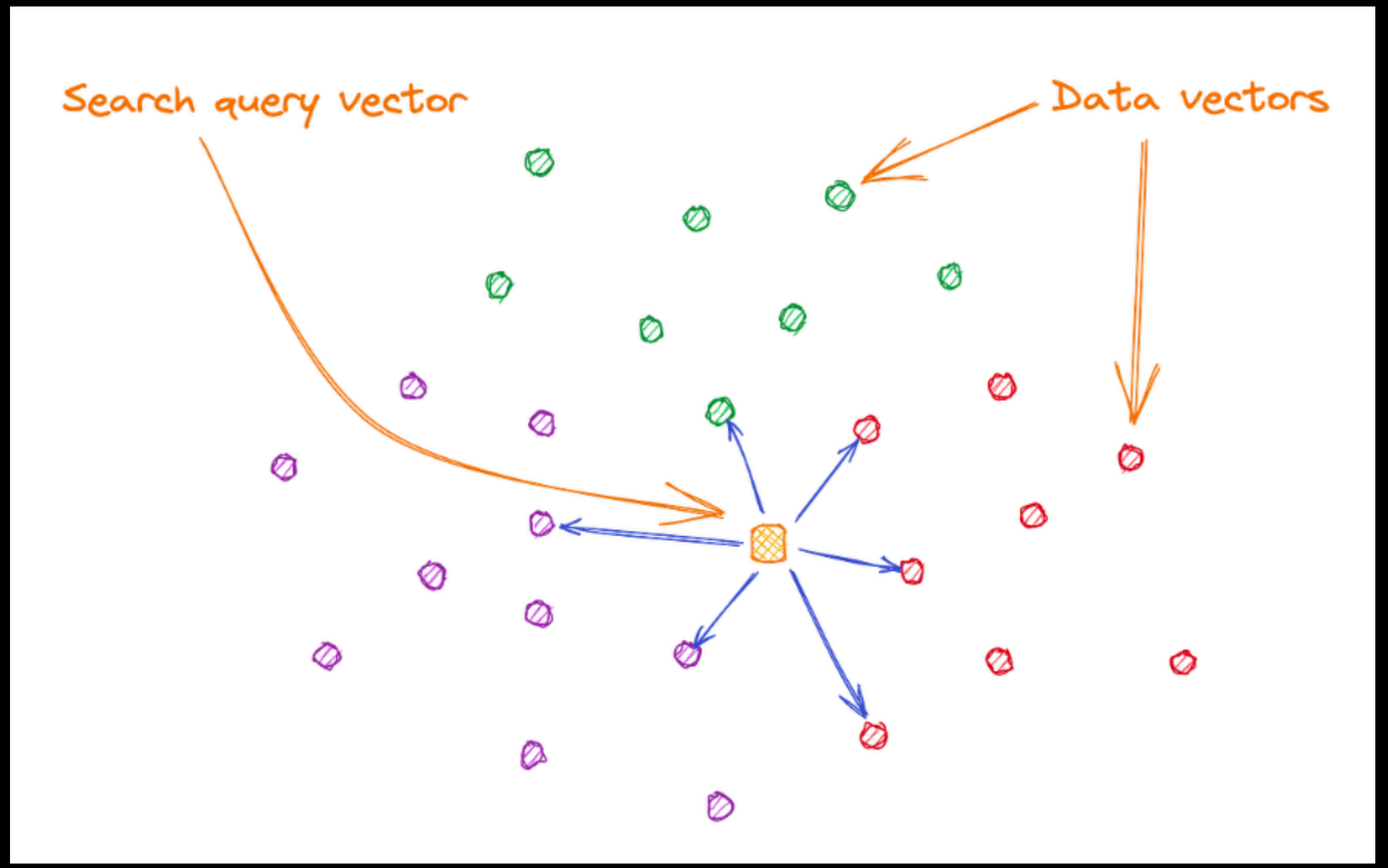
Embedings Representation in VectorDB





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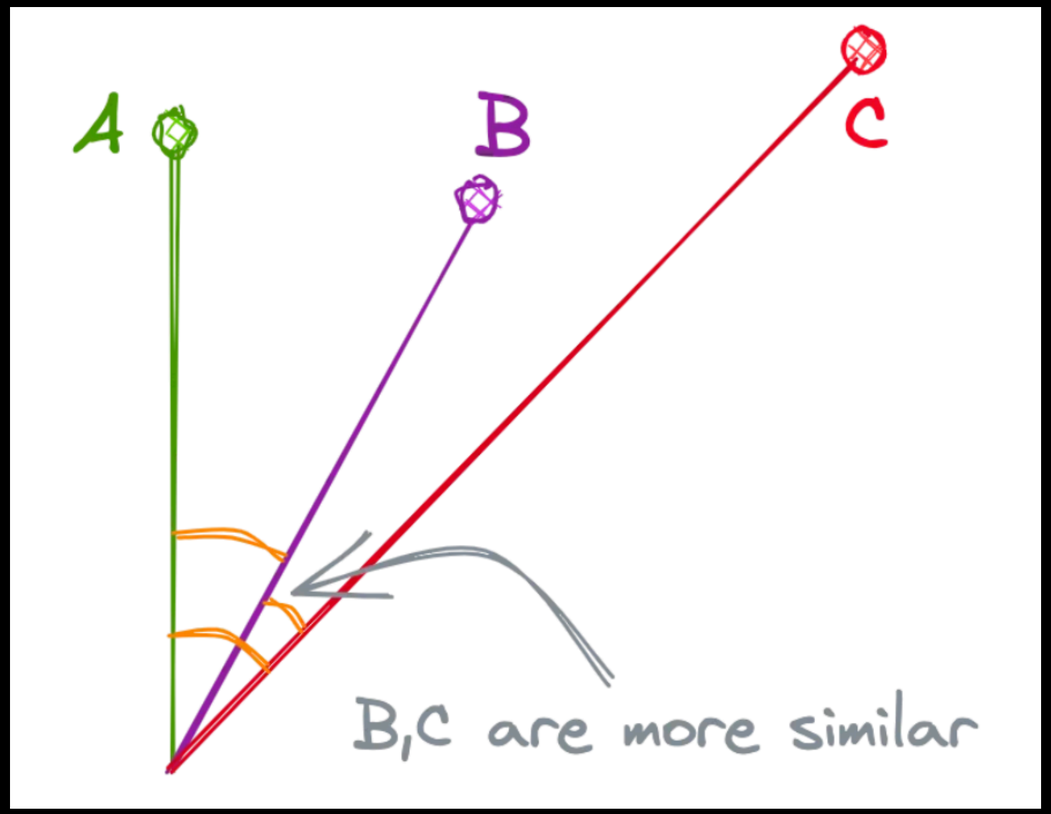
Similarity Search Algorithm

There are few similarity search algorithms but Cosine Similarity is best



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Optimizing Similarity Search

Most of the VectorDB uses Hierarchical Navigable Small World graph (HNSW) algorithm for better similarity search.

Why do we need Optimizing Similarity Search?

If there are one million data vectors, there will be one million similarity calculations, and the most similar data vectors will be returned as the result. However, when dealing with millions or billions of records, this can severely impact performance due to the large number of calculations and memory required.

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Hierarchical Navigable Small World graph (HNSW)

This optimization combines two different methods:

1. Probability Skip List
2. Navigable Small World Graph.

It creates a multi-layer graph from the given data, which helps the algorithm quickly find approximate nearest neighbors.

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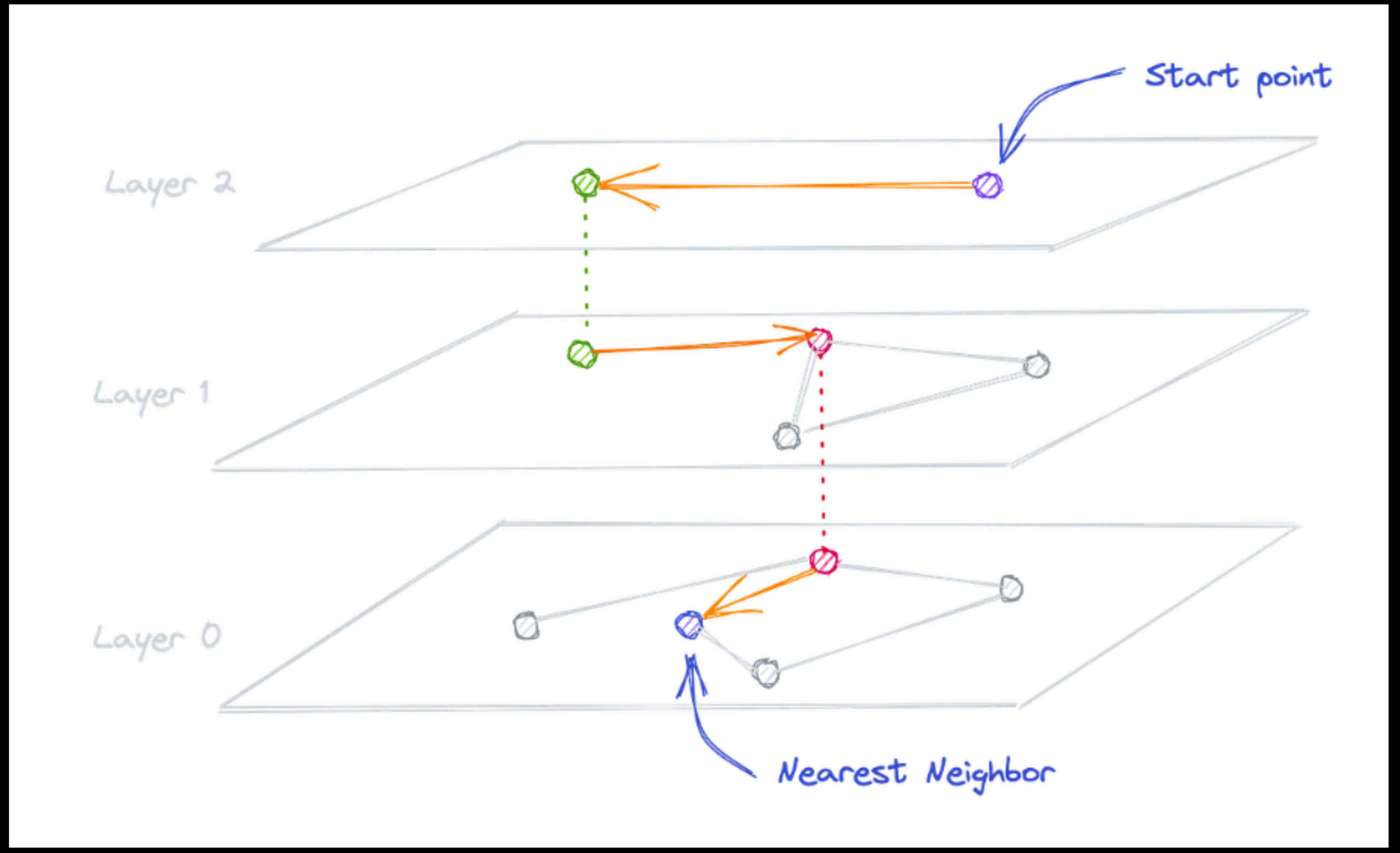
Hierarchical Navigable Small World graph (HNSW)



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Benifits Of VectorDB Over Traditional Databases

There are two main benifits of using VectorDB:

- Fast Retrieval
- Optimized Storage

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ChromaDB - Create Instance

```python
from langchain_chroma import Chroma

vector_store = Chroma(
    collection_name="example_collection",
    embedding_function=embeddings,
    # Where to save data locally, remove if not necessary
    persist_directory="./chroma_langchain_db",
)
```

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CRUD Operations on VectorDB

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ChromaDB - Add Documents → 1

```python
from uuid import uuid4
from langchain_core.documents import Document

document_1 = Document(
    page_content=("I had chocolate chip pancakes and "
    "scrambled eggs for breakfast this morning."),
    metadata={"source": "tweet"},
    id=1,
)
document_2 = Document(
    page_content=("The weather forecast for tomorrow "
    "is cloudy and overcast, with a high of 62 degrees."),
    metadata={"source": "news"},
    id=2,
)
```

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ChromaDB - Add Documents → 2

```
documents = [
    document_1,
    document_2
]
uuids = [str(uuid4()) for _ in range(len(documents))]
vector_store.add_documents(documents=documents, ids=uuids)
```

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ChromaDB - Update Documents → 1

```
updated_document_1 = Document(
    page_content=("I had chocolate chip pancakes "
    "and fried eggs for breakfast this morning."),
    metadata={"source": "tweet"},
    id=1,
)
updated_document_2 = Document(
    page_content=("The weather forecast for tomorrow "
    "is sunny and warm, with a high of 82 degrees."),
    metadata={"source": "news"},
    id=2,
)
```

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ChromaDB - Update Documents → 1

```
# Update single document
vector_store.update_document(document_id=uuids[0], document=updated_document_1)

# You can also update multiple documents at once
vector_store.update_documents(
    ids=uuids[:2], documents=[updated_document_1, updated_document_2]
)
```

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ChromaDB - Delete Documents

```
updated_document_1 = Document(
    page_content=("I had chocolate chip pancakes "
    "and fried eggs for breakfast this morning."),
    metadata={"source": "tweet"},
    id=1,
)


vector_store.delete(ids=[1])
```

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ChromaDB - Search Upon Query

```
results = vector_store.similarity_search(
    "LangChain provides abstractions to make working with LLMs easy",
    k=2,
    filter={"source": "tweet"},
)
for res in results:
    print(f"* {res.page_content} [{res.metadata}]")
```

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ChromaDB - Search Upon Query with Score

```
results = vector_store.similarity_search_with_score(
    "Will it be hot tomorrow?", k=1, filter={"source": "news"}
)
for res, score in results:
    print(f"* [SIM={score:3f}] {res.page_content} [{res.metadata}]")
```

To Learn More About VectorDBs Integeration in Langchain Visit:
https://docs.langchain.com/oss/python/integrations/vectorstores

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Exercise

Do this exercise using an open-source embedding model from Hugging Face. Use different documents and queries to test similarity search.

1. Choose a small open-source embedding model (e.g., "sentence-transformers/all-MiniLM-L6-v2").
2. Create a new set of documents (e.g., Pakistani cities, historical places, or favorite sports).
3. Write a query and find the most relevant document using cosine similarity.
4. Print the query, top document, and similarity score.

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Use Case: Smart Product Assistant

Goal: An AI chatbot that answers user questions directly from official product manuals using Retrieval-Augmented Generation (RAG).

Purpose and benefit: Enhance user experience

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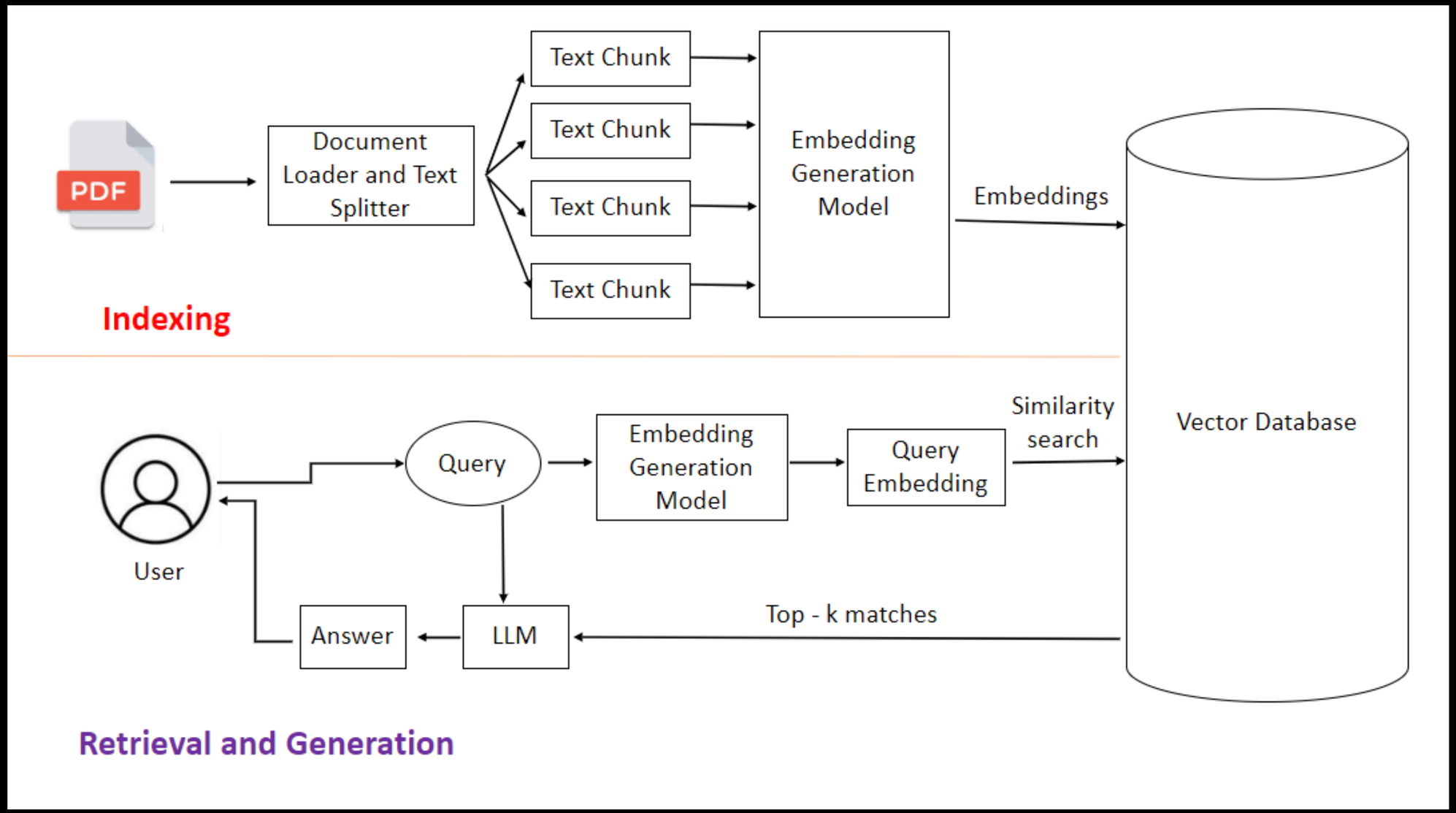
Use Case: Architecture Diagram



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Tools

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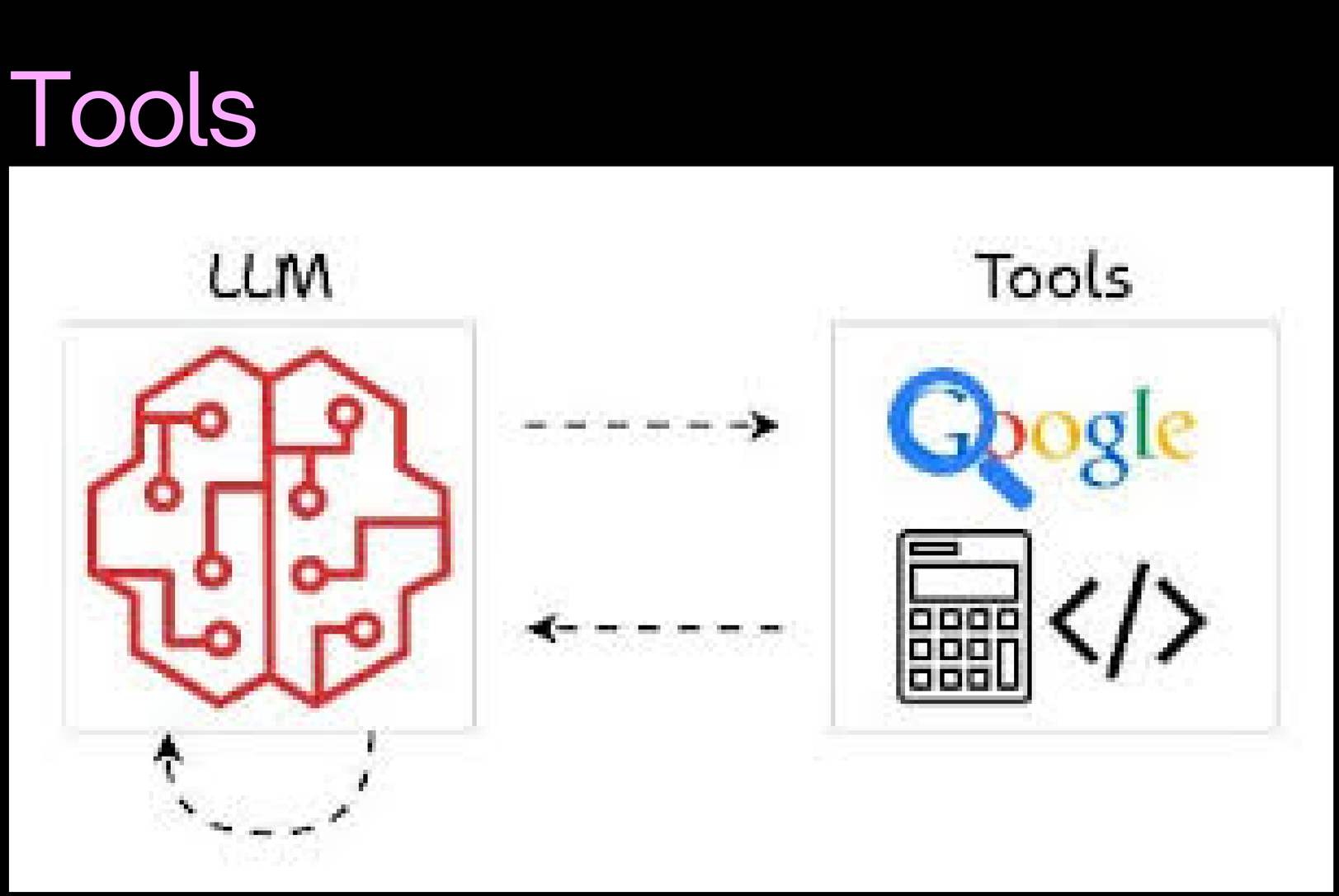
What is Tool?

A tool is just a python function that LLM can understand and call when needed

Why do we need Tools?

As LLMs can’t perform actions so, tools enable LLMs to perform actions

Tools



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Types of Tools in Langchain

There are two kinds of tools support in langchain, as follows:

- Built-in Tool
- Custom Tool

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Built-in Tool

LangChain provides built-in tools to let LLMs interact with real systems.

Some of them are below:

- DuckDuckGo
- ShellTool
- PythonREPLTool

Explore more tools:
https://docs.langchain.com/oss/python/integrations/tools

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Built-in Tool - DuckDuckGo Search

```
from langchain_community.tools import DuckDuckGoSearchRun

search_tool = DuckDuckGoSearchRun()

results = search_tool.invoke('top news in pakistan today')

print(results)
```

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Built-in Tool - DuckDuckGo Search - Multiple Results

```python
from langchain_community.tools import DuckDuckGoSearchResults

search_tool = DuckDuckGoSearchResults(num_results=5)
results = search_tool.invoke("Top today AI News")

entries = results.split("snippet: ")[1:]

for i, entry in enumerate(entries, start=1):
    parts = entry.split(", title: ")
    snippet = parts[0].strip()
    title, link = parts[1].split(", link: ")

    print(f"\nResult {i}")
    print("Title:", title.strip())
    print("Snippet:", snippet.strip())
    print("Link:", link.strip())
```

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Built-in Tool - ShellTool

```
from langchain_community.tools import ShellTool

shell_tool = ShellTool()

results = shell_tool.invoke('ls')

print(results)
```

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Built-in Tool - PythonREPLTool

```
from langchain_experimental.tools import PythonREPLTool
tool = PythonREPLTool()

result = tool.run("print(2 + 3)")
print(result)
```

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Custom Tool

A custom tool is a tool that you define yourself

When you will move to custom tool option?

- If you want to call your own API
- If you want to make your custom business logic

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Custom Tool - Tool Decorator

```python
from langchain_core.tools import tool
@tool
def multiply(a: int, b:int) -> int:
    """Multiply two numbers"""
    return a*b
```

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Custom Tool - StructuredTool → 1

```python
from langchain_core.tools import StructuredTool
from pydantic import BaseModel, Field

class MultiplyInput(BaseModel):
    a: int = Field(required=True, description="The first number to multiply")
    b: int = Field(required=True, description="The second number to multiply")

def multiply_func(a: int, b: int) -> int:
    return a * b
```

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Custom Tool - StructuredTool → 2

```
multiply_tool = StructuredTool.from_function(
    func=multiply_func,
    name="multiply",
    description="Multiply two numbers",
    args_schema=MultiplyInput
)
result = multiply_tool.invoke({'a':3, 'b':5})
```

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Custom Tool - StructuredTool → 3

```
print(result)
print(multiply_tool.name)
print(multiply_tool.description)
print(multiply_tool.args)
```

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Tool-Augmented Chatbot for Imtiaz Super Store

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Use Case: E-Commerce Personal Assistant

Imtiaz Super Store wants to build an **AI-powered shopping assistant** that goes beyond chat and performs **real business operations** using live product data and customer loyalty information.
This chatbot acts as a **personal shopping assistant**, helping users:

- Discover products
- Apply loyalty points intelligently
- Get final, real-time pricing
- Check their loyalty balance

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End of Class

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